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Oefeningenreeks 2E nr. 5.9

$$
\begin{aligned}
&\lim_{x\to 0}\left(\frac{1+3x}{1-x}\right)^{\frac{3}{x}} = \lim_{x\to 0}\left(\frac{1+4x-x}{1-x}\right)^{\frac{3}{x}}\\
&= \lim_{x\to 0}\left(\frac{1-x}{1-x}+\frac{4x}{1-x}\right)^{\frac{3}{x}} = \lim_{x\to 0}\left(1+\frac{4x}{1-x}\right)^{\frac{3}{x}\cdot\frac{1-x}{4x}\cdot\frac{4x}{1-x}}\\
&= \lim_{x\to 0}\left(\left(1+\frac{4x}{1-x}\right)^{\frac{1-x}{4x}}\right)^{\frac{12x}{x(1-x)}} = \lim_{x\to 0} e^{\frac{12x}{x-x^2}}\\
&= \lim_{x\to 0} e^{\frac{12}{1-x}} = e^{\frac{12}{1-0}} = e^{12}
\end{aligned}
$$

References